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192312523

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I K. Reddi Kishore Reddy (192312523) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding, major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

ASSESSMENT-CO1-AT3 :-

COURSE CODE :- CSA1714

COURSE NAME :- Artificial intelligence

Name :- K.Reddi Kishore Reddy

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Smart Home Automation system using intelligent Agents.

Introduction:

A smart home automation system uses Artificial Intelligence (AI) to automatically control lighting, temperature, and security based on sensor data, user behavior, and environmental conditions. The system receives real-time inputs from motion sensors, temperature sensors, cameras, and clocks, while also learning the user's daily habits to improve comfort, energy efficiency, and security.

@ simple Reflex Agent

Definition:

A simple Reflex Agent makes decisions based only on the current input (Percept) using Predefined IF-THEN rules. It does not remember Previous events.

working in Smart Home:

* If motion is detected → Turn ON the lights.
* If no motion is detected for 10 minutes → Turn OFF the lights.
* If room temperature exceeds 30°C - Turn ON the air conditioner.
* If smoke is detected → Activate the fire alarm.

<u>Advantages:</u>
* very fast response.
* Easy to design and implement.
* suitable for simple automation taks.

<u>Disadvantages:</u>
* Has no memory of previous events.
* cannot adapt to changing situations.
* cannot learn user preferences.

(b) <u>Model - Based</u> Reflex Agent.

<u>Definition:</u>

A model - based Agent maintains an internal model (memory) of the environment. it combines current sensor data with previous information befor making decisions.

Working in the smart Home:-

* Remembers whether a room was occupied recently
* Detects whether lights are already ON or OFF
* Tracks indoor and outdoor temperatures.
* monitors door and window status for security

Advantages:-

* works well in Partially observable environments.
* Avoids unnecessary repeated actions.
* makes more accurate decisions.

Disadvantages:-

* Requires additional memory.
* more complex than a simple reflex agent.

© Goal-Based Agent.

Definition:-

A Goal-Based Agent chooses actions that help achieve specific goals using planning and search techniques.

Goals in the smart Home:-

* Maintain room temperature at 24°C.

* Keep the house secure.
* Reduce electricity consumption.

working:-

Example:
* Goal: Maintain comfortable temperature.
* current temperature: 32°C.
* Action: Turn ON the air conditioner until the room reaches 24°.

Advantages:-
* makes intelligent decisions.
* can plan multiple actions.

Disadvantages:-
* Requires planning algorithms.
* Takes more processing time.

④ utility - Based Agent.

Definition:-

A utility - Based Agent selects the action that provides the highest overall benefit (utility) by considering multiple objectives simultaneously.

working in the smart Home:-

the agent balances:
* user comfort.

* Energy consumption.

* security.
* Electricity cost

Advantages:-

* Produces optimal decisions.
* Handles conflicting objectives.

Disadvantages:-

* complex to design
* Requires utility calculations.

2. Robot Navigation in an unknown maza using uninformed search.

Introduction:-

A robot is placed in an unknown maza without any prior knowledge of the layout. Its objective is to find a path from the starting position to the exit. since no map or heuristic information is available, the robot uses uninformed search algorithms such as uniform cost search (UCS) and Iterative Deepening search (IDS)

(a) uniform cost search (UCS)

Definition:-

uniform cost search expands the node with the lowest cumulative path cost first. It always selects the least expensive path discovered so far.

working:-

1. start from the initial position.

2. calculate the cumulative cost of each possible move.

3. Expand the node with the smallest total cost.

4. continue until the exit is reached.

Advantages:-

* complete (always finds a solution if one exists)
* finds the optimal (lowest-cost) path.
* suitable for variable movement costs.

Disadvantages:-

* High memory usage.
* slow for very large search spaces.

(b) iterative Deepening search (IDS)

Definition:-

Iterative Deepening search repeatedly performs Depth-first search with increasing depth limits untill the goal is found.

Working:-

* Depth Limit = 0 → Search only the start node.
* Depth Limit = 1 → search one level deeper.
* Depth Limit = 2 → search two levels deeper.
* continue increasing the depth until the exit is found.

Advantages:-

* Requires very little memory.
* complete for finite search spaces.
* suitable when the goal depth is unknown.

Disadvantages:-

* Repeats searches at each depth.
* Does not guarantee the least-cost path when step costs differ.

complexity

* Time complexity: $O(b^d)$
* space complexity: $O(b \times d)$.

* b = branching factor.
* d = depth of the goal node.

Intelligent Agents in maze navigation.

Simple Reflex Agent:

* Responds only to immediate surroundings:
* May revisit the same locations.
* can become trapped indoors.

Model-Based Agent:-

* maintains an internal map of explored areas.
* Remembers visited paths.
* Avoids repeated exploration.

Conclusion:-

* Smart Home Automation: A Hybrid agent is the best choice because it adapts to user preference while optimized and security.
* maze Navigation: A Model-Based Goal-Based Agent using uniform cost search (UCS) provides the most effective solution in uniform.